PORTFOLIO PACKAGE · HEALTHCARE ADMINISTRATION

AI in U.S. Hospital Revenue Cycle Management

How AI is changing claims, denials and prior authorization, and where health systems, community hospitals and billing companies should start.

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Independent research project · September 2026

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PROJECT AT A GLANCE

AI delivers targeted wins in the revenue cycle, not yet a new cost structure

This project assesses how AI is changing U.S. hospital revenue cycle management, with a focus on claims, denials and prior authorization, and builds a practical adoption case for three types of organizations.

SCOPE

United States · revenue cycle and claims (billing, coding, prior authorization, denials)

METHOD

Synthesis of 28 industry, government and trade sources (2021–2026) plus an illustrative ROI model

OUTPUT

Research report, 15-slide executive deck, three-scenario business case

Key findings

- **The problem is rework.** Providers spent $25.7B fighting claim denials in 2023. About 15% of private-payer claims are denied at first, and roughly 70% of those are eventually paid.
- **Interest is broad; scale is thin.** 80% of health systems are exploring, piloting or implementing generative AI for the revenue cycle, but only 27% run AI at scale.
- **Published gains are real but narrow.** Case studies report 18–71% improvements on single workflows such as claim-status follow-up, days in A/R and prior-auth turnaround.
- **Insurers are automating faster.** More than half of health plans use AI in administrative work, versus 25% of providers, fueling an "AI arms race" over coding, denials and appeals.
- **Regulation is tightening.** Federal prior-auth decision deadlines took effect in January 2026, payer APIs are due January 2027, and at least eight states bar algorithm-only denials.
- **The business case holds across settings.** A modeled Phase 1 rollout returns $2.5M–$22.3M in first-year net benefit and frees the work of 11–85 full-time staff.

Deliverables

Deliverable	What's inside	Formats	In this PDF
Research report	8 sections; case-study chart, rollout roadmap, cost-savings and time-savings charts; 28 linked sources	Live doc · PDF · Word	pp. 4–19
Executive slide deck	15 slides with speaker notes, from the denial problem to the business case	Live deck · PDF · PowerPoint	pp. 20–34
Three-scenario ROI model	Major health system, 300-bed community hospital, billing company; sensitivity analysis	Report §7 · slides 11–13	p. 3
Resume bullets	Full (6 bullets) and short (3 bullets) versions	Text	p. 35

Live versions: research report · slide deck. The live links open only for people the author has shared them with.

BUSINESS CASE AT A GLANCE

All three cases pay back in year one, for different reasons

The model prices one Phase 1 bundle (denial prevention plus claim-status automation) for three hypothetical organizations. Every input is an assumption to replace with local data.

	Major health system	300-bed community hospital	Billing company
Net first-year benefit	**$22.3M**	**$2.5M**	**$3.9M**
Net benefit vs. AI cost	4.5x	4.2x	2.0x
Main source of value	Recovered revenue (79%)	Recovered revenue (73%)	Staff time (51%)
Denials avoided per year	60,000	9,600	66,000
Staff hours freed per year	~150k (72 FTEs)	~24k (11 FTEs)	177k (85 FTEs)
Most sensitive to	Denial reduction: at 10%, net ≈ $14.0M	Claim value: at $400, net ≈ $1.4M	Hours freed: at half, net ≈ $2.4M

How the model works

Net benefit = denials avoided × (rework cost + share never recovered × average claim value × share of revenue kept) + staff hours freed × hourly cost − annual AI cost

Key inputs	Major health system	300-bed hospital	Billing company
Claims per year	4,000,000	400,000	3,000,000
Initial denial rate / AI reduction	10% / 15%	12% / 20%	11% / 20%
Rework cost per denied claim	$57.23	$57.23	$35
Average written-off claim / share kept	$1,200 / 100%	$800 / 100%	$500 / 6% fee
Staff hours freed (automation) / hourly cost	60,000 / $38	8,000 / $35	100,000 / $30
Annual AI cost, incl. integration	$5.0M	$0.6M	$2.0M

Bases: Premier (2025) for the $57.23 rework cost and ~70% recovery rate; Tebra (2025) for billing fees of 4–10% of collections; BLS (May 2025) for the $24.59 median hourly wage. Other inputs are assumptions. Full detail in the report's implementation playbook and slides 11–13.

Recommended rollout

PHASE 1 · MONTHS 0–6

Automate rote work: claim-status follow-up, eligibility checks, claim scrubbing

PHASE 2 · MONTHS 6–18

Assist skilled staff: coding assistance and CDI, prior-auth packets, appeal drafting

PHASE 3 · MONTHS 18–36

Scale autonomy: autonomous coding, API-based prior auth, patient billing assistant

AI in U.S. Hospital Revenue Cycle Management

Sep 25, 2026 · @Neha

Executive summary

AI is moving from pilot to production in U.S. hospital revenue cycles, and it pays off most where work is high-volume and rules-based. The gains are real but uneven, and insurers are deploying AI just as fast.

- **The problem is large and growing.** Providers spent $25.7 billion fighting claim denials in 2023, up 23% in a year. About 15% of private-payer claims are denied at first, yet roughly 70% of those are eventually paid.
- **Interest is near-universal; scale is not.** 80% of health systems are exploring, piloting or implementing generative AI for the revenue cycle. Only 27% run AI at scale across multiple functions.
- **The best-documented wins are in coding and claims follow-up.** One system halved accounts-receivable days from 14 to 7 with autonomous coding. Another removed 71% of claim-status accounts from staff queues and saved 5,559 work hours in eight months.
- **Payers are automating too.** 84% of insurers surveyed use AI or machine learning, and providers now describe an "AI arms race" over coding, denials and appeals.
- **Regulation is catching up.** Federal prior-authorization deadlines took effect in January 2026, and at least eight states now bar algorithms from being the sole basis for a denial.

Recommendation: start with "no-regret" use cases (claim status, eligibility, coding assistance, appeal drafting), measure them against a baseline, and keep a human accountable for every payer-facing decision.

Why the revenue cycle is ripe for AI

The U.S. revenue cycle is a chain of repetitive, data-heavy handoffs between hospitals and payers, and each handoff is a chance for a claim to fail. Administration consumes about one quarter of the roughly $4 trillion the U.S. spends on health care. McKinsey estimates up to $265 billion a year of that could be saved.

The revenue cycle has three stages, and AI touches all of them:

- **Front end** – scheduling, registration, insurance eligibility, prior authorization.
- **Mid-cycle** – clinical documentation, coding, charge capture.
- **Back end** – claim submission, payment posting, denial management, appeals, patient billing.

The denial problem in numbers

Metric	Value	Source
Private-payer claims denied on first submission	~15%	Premier, 2025
Denied claims ultimately paid after rework	~70%	Premier, 2025
Provider cost to adjudicate one claim (2023)	$57.23	Premier, 2025
Total provider cost of fighting denials (2023)	$25.7B, of which ~$18B potentially unnecessary	Premier, 2025
Providers with denial rates of 10% or more	41%	Experian Health, 2025
Medicare Advantage prior-auth requests denied (2024)	7.7% of 52.8M	KFF, 2026
Appealed MA prior-auth denials overturned (2024)	80.7%	KFF, 2026
Physician and staff time on prior auth	~13 hours/week	AMA, 2026

The pattern matters more than any single number: most denials are eventually reversed, so the cost lies in rework, not lost care. That is exactly the kind of waste automation targets.

Root causes are mostly data problems

Experian's 2025 survey found the top denial causes were missing or inaccurate data (50%), authorizations (35%) and registration errors (32%). Meanwhile, 43% of respondents said they were understaffed. Error-prone data plus thin staffing is the gap AI tools are being bought to close.

Where AI is applied

AI in the revenue cycle falls into three technical families: robotic process automation for fixed clicks, machine learning for prediction, and generative or "agentic" AI for reading and writing text. Most hospitals combine all three.

Stage	Use case	What the AI does	Maturity (2026)
Front end	Eligibility and coverage discovery	Checks active coverage and finds unbilled insurance before the visit	Mature
Front end	Prior authorization	Pulls clinical evidence from the record, fills payer forms, tracks status	Scaling
Front end	Patient estimates and financial clearance	Predicts out-of-pocket cost and propensity to pay	Scaling
Mid-cycle	Ambient documentation	Drafts the clinical note from the visit conversation	Scaling fast
Mid-cycle	Clinical documentation improvement (CDI)	Flags missing diagnoses or vague documentation for query	Scaling
Mid-cycle	Autonomous or assisted coding	Assigns ICD-10/CPT codes and routes only uncertain charts to coders	Scaling
Back end	Claim scrubbing and denial prediction	Scores each claim for denial risk and fixes errors before submission	Mature

Stage	Use case	What the AI does	Maturity (2026)
Back end	Claim-status follow-up	Queries payer portals and resolves routine status work without staff	Mature
Back end	Appeal letters	Drafts appeals from the denial code and the chart	Emerging
Back end	Patient billing service	Answers billing questions and sets up payment plans	Emerging

Oliver Wyman's 2026 survey found 92% of leaders agree some AI investments are "no-regret." The four named most often were ambient documentation, CDI, coding automation and electronic prior authorization.

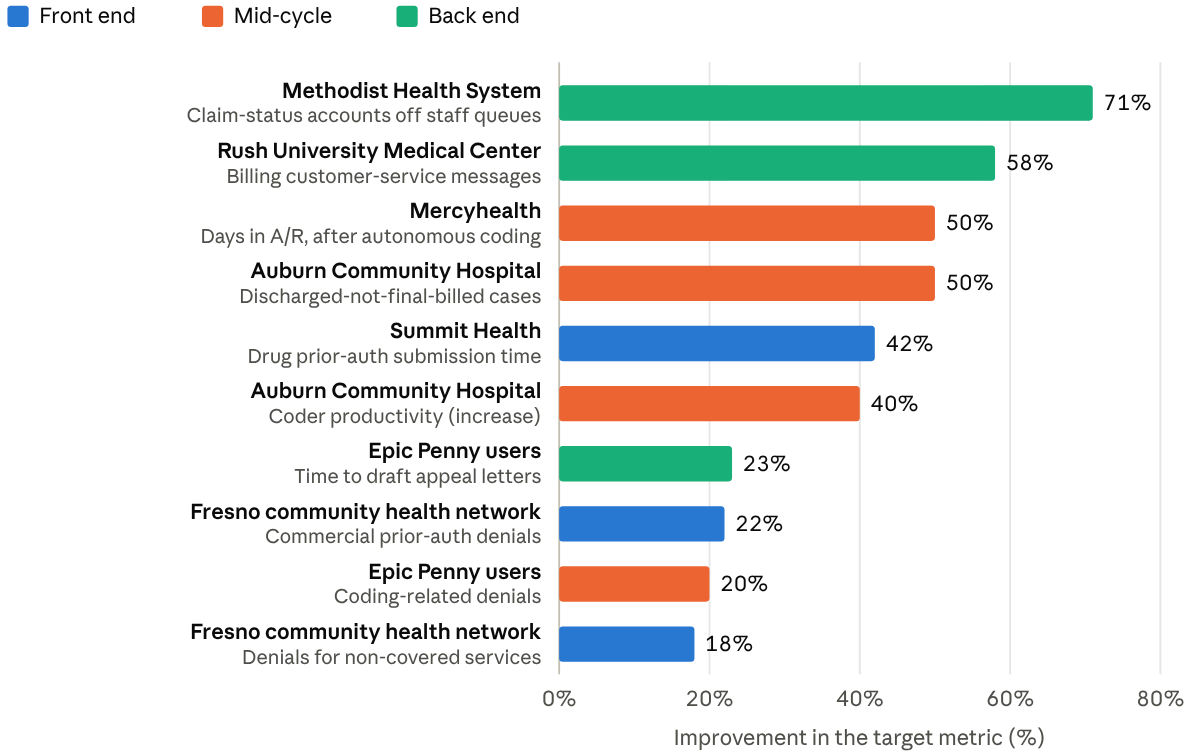
The EHR is becoming the delivery channel. Epic's revenue-cycle agent, Penny, now helps draft appeals and flag coding issues inside the system staff already use (Epic), which lowers the integration barrier for mid-sized hospitals.

Evidence of impact

Where hospitals have published results, AI typically improves one targeted workflow by 20–50%, and the largest gains come in rote back-office work. Enterprise-wide financial impact is still rarely reported.

Published case studies report 18–71% gains on single workflows

Reported improvement in each target metric (%), as stated by the provider or vendor



Compiled from AHA, Epic, Fierce Healthcare and Becker's case studies, 2024–2026

Case notes

Organization	Setting	Tool or approach	Other reported results	Source
Methodist Health System	Health system	AKASA claim-status automation	56,118 accounts resolved and 5,559 staff hours saved in 8 months	Fierce Healthcare
Mercyhealth	$5B system, 85 locations	Arintra autonomous coding	88% of charts sent straight to billing; 5.1% revenue uplift	Becker's

Organization	Setting	Tool or approach	Other reported results	Source
Auburn Community Hospital (NY)	99-bed rural hospital	RPA, natural-language processing and machine learning, built over ~10 years	Case mix index up 4.6%	AHA
Fresno community health network (CA)	Community network	AI review of claims before submission	30–35 staff hours a week saved on appeals, with no new hires	AHA
Summit Health	Multispecialty group	Epic AI for medication prior authorization	—	Epic
Rush University Medical Center	Academic medical center	Epic Emmie patient-facing assistant	—	Epic
Epic Penny users	Multiple Epic customers	Penny generative AI for coding and appeals	Results from the most active users	Epic

Adoption is broad but shallow

- **Interest:** 80% of health systems are exploring, piloting or implementing generative AI for revenue cycle, up from 58% considering it in 2023 (HFMA–AKASA via Fierce Healthcare, n=519).
- **Scale:** only 27% deploy AI at scale across multiple functions; 53% are still in pilots (HFMA, 2026, n=95).
- **Size gap:** 64% of large systems are piloting or implementing, versus about 20% of systems with $500M–$1B in revenue.
- **Claims specifically:** just 14% of providers use AI for claims, but 69% of those users report fewer denials or better resubmission success (Experian Health, 2025).

Read these results with care

Nearly all published outcomes come from vendors or early-adopter hospitals, without control groups or independent audit. Projections such as McKinsey's 30–60% cut in cost to collect (via HFMA) are estimates, not measured results. A strong portfolio or business case should treat them as hypotheses to test locally.

The payer side: an AI arms race

Insurers adopted AI earlier and more widely than hospitals, so the revenue cycle is becoming a contest between two sets of algorithms. More than half of health plans use AI in administrative workflows, versus 25% of providers (CAQH Index, 2026).

Battleground	Payer use of AI	Provider response
Prior authorization	Automated approvals; Aetna says 80% of its prior auths are processed in real time	AI assembles clinical evidence, submits requests and tracks status
Coding	Automated "downcoding" edits and coding-intensity analytics	Ambient scribes, CDI prompts and autonomous coding
Claim adjudication	Every claim screened before payment, not samples; Aetna's ACE tool flags gaps and inconsistencies	Pre-submission scrubbing and denial prediction
Denials and appeals	Algorithms that predict length of post-acute care (alleged to drive denials)	Appeal letters drafted automatically from the chart and denial code

Sources: Healthcare Finance News, Healthcare Brew, AHA.

Three flashpoints

1. **How insurers use AI.** A NAIC survey found 84% of 93 insurers use AI or machine learning, 68% for prior-auth approvals; only 12% said they use or consider it to deny requests. Insurers stress that physicians make every denial. Aetna's Nathan Frank: "We never use AI to deny prior authorizations."

2. **AI scribes and coding intensity.** Blue Health Intelligence attributes about $2.3 billion in added spending to AI scribes. STAT reports both sides privately agree scribes raise coding intensity. The AHA counters that sicker patients explain the rise, citing a ~5% case-mix increase from 2019 to 2024.
3. **Litigation.** A class action over UnitedHealth's nH Predict algorithm is proceeding on breach-of-contract claims. Plaintiffs allege nine in ten appealed denials were reversed; UnitedHealth says the tool only guides care planning.

What it means for hospitals

An arms race raises costs on both sides without improving care, and smaller providers fall behind first. "The smaller provider is not going to be able to catch this stuff fast enough," said PYA's Valerie Rock. The more durable path is shared rails: insurers in the June 2025 AHIP pledge cut prior-auth volume 11% (6.5 million requests) by April 2026.

Risks, regulation and ethics

Regulators are pushing prior authorization toward faster, electronic and human-accountable decisions, while hospitals carry their own risks from AI errors, compliance exposure and vendor concentration.

Regulatory milestones

Date	Rule or event	What it means for hospitals
Jan 1, 2027	CMS-0057-F API deadline	Impacted payers must run FHIR Prior Authorization, Provider Access and Payer-to-Payer APIs, enabling end-to-end electronic prior auth
Apr 2026	KFF state tracker	At least eight states, including California, Illinois, Maryland and Texas, bar algorithms from being the sole basis for a denial
Mar 31, 2026	CMS-0057-F public reporting	Payers publish 2025 approval, denial, appeal and turnaround metrics, giving hospitals benchmarks for negotiation

Date	Rule or event	What it means for hospitals
Jan 1, 2026	CMS-0057-F decision deadlines	Medicare Advantage, Medicaid and CHIP plans (not exchange plans) must decide in 72 hours (expedited) or 7 days (standard)
Jan 1, 2026	WISeR model launch	AI-assisted prior auth comes to traditional Medicare in six states for six years; vendors earn a share of savings from denials
2024	CMS Medicare Advantage guidance	Medical-necessity denials must reflect individual patient circumstances, not an algorithm alone

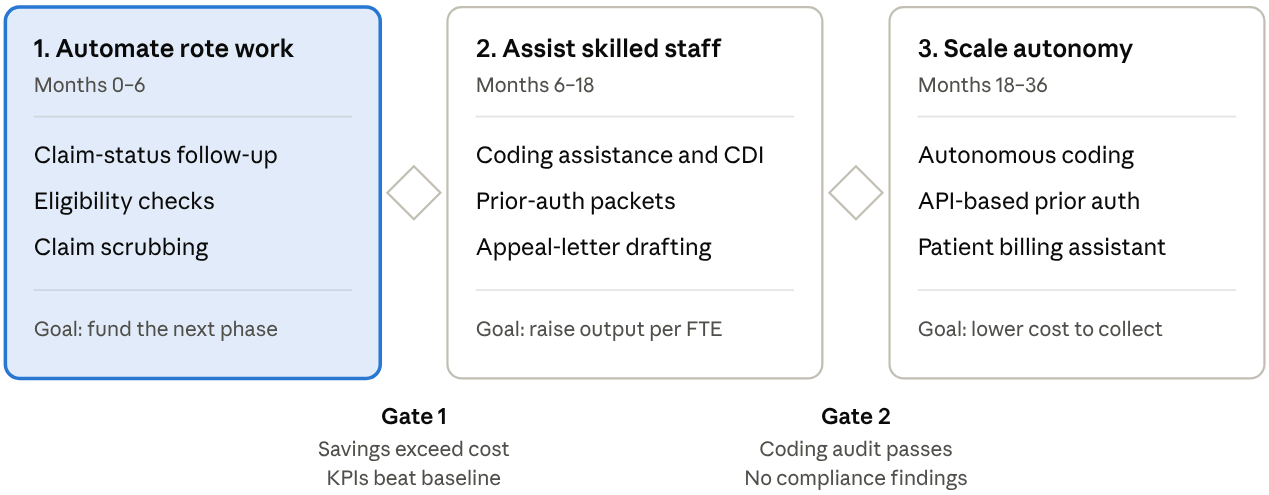
Risks hospitals must manage

- **Errors at scale.** 25% of finance leaders fear "AI errors could be scaled faster than humans can catch" (HFMA, 2026). A wrong coding rule applied to 10,000 claims is a bigger problem than one coder's mistake.
- **Compliance and audit exposure.** Coding tools that raise acuity without documentation support invite payer clawbacks and False Claims Act risk. Every AI-suggested code needs a documentation trail.
- **Automation bias.** A Stanford review of 21 utilization-management tools in *Health Affairs* flagged anchoring on AI output and found no studies comparing wrongful-denial rates with and without AI (Becker's).
- **Vendor and cyber concentration.** The 2024 Change Healthcare ransomware attack exposed data on about 192.7 million people and froze claims flow nationwide. More AI vendors in the claims path means more single points of failure.
- **Workforce.** Only 7.4% of leaders call their workforce "very prepared" for new skills, and 21% cite disruption or morale. Mayo Clinic's framing is to "elevate, not eliminate" revenue-cycle roles (HFMA).
- **Equity between hospitals.** Large systems pilot AI at roughly three times the rate of smaller ones, so rural and safety-net hospitals risk losing ground in the arms race.

Implementation playbook

A hospital should sequence AI from rote, easily measured back-office tasks toward autonomy, and let each phase's savings fund the next. This is a recommended approach, not a standard.

Three phases move from rote automation to autonomy, gated by results



Recommended roadmap · 3 phases, 2 gates

Phase 1 tools are the most mature and the easiest to measure. Phase 3 waits for payer APIs required by January 2027 and for audited proof that coding accuracy holds.

Build the business case

The model below prices one Phase 1 bundle, denial prevention plus claim-status automation, for a hypothetical 300-bed community hospital. It then adds two comparison cases: a major health system and a third-party billing company. Every input is an assumption to replace with local data.

Net benefit = (C × d × r) × (k + w × v × f) + H × h − S

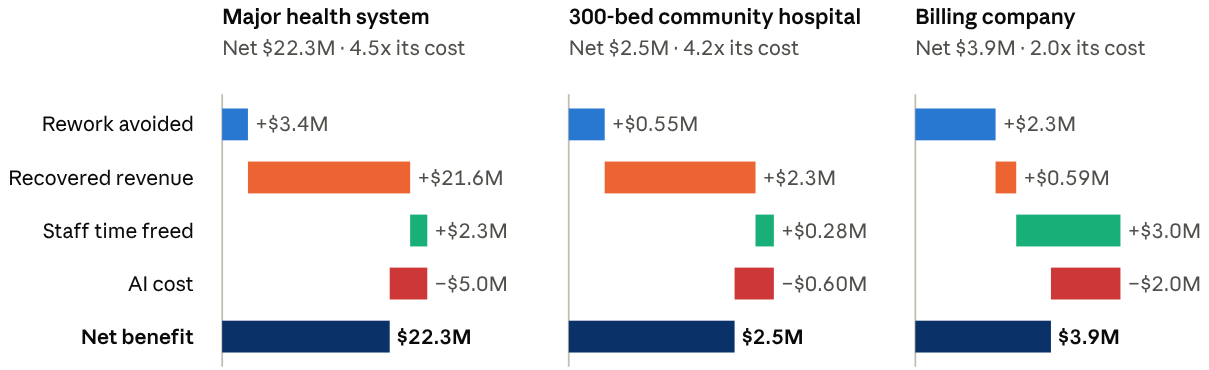
Input	Symbol	Major health system	300-bed community hospital	Billing company	Basis
Claims per year	C	4,000,000	400,000	3,000,000	Assumption
Initial denial rate	d	10%	12%	11%	Below the ~15% average (Premier); large systems assumed more mature

Input	Symbol	Major health system	300-bed community hospital	Billing company	Basis
Reduction in denials from AI	r	15%	20%	20%	Low end of published case studies; less headroom at a mature system
Rework cost per denied claim	k	$57.23	$57.23	$35	Premier for providers; lower-cost labor assumed at the billing firm
Share of denials never recovered	w	30%	30%	30%	Inverse of the ~70% eventually paid
Average value of a written-off claim	v	$1,200	$800	$500	Assumption; higher acuity at the large system, physician-heavy mix at the billing firm
Share of recovered revenue kept	f	100%	100%	6%	Billing firms typically charge 4–10% of collections (Tebra)
Staff hours freed per year	H	60,000	8,000	100,000	Community figure annualized from Methodist's 5,559 hours in 8 months; billing firm = 25% of ~200 FTEs
Loaded cost per staff hour	h	$38	$35	$30	1.2–1.5x the $24.59 median wage for medical records specialists (BLS)
Annual AI cost, incl. integration	S	$5.0M	$0.6M	$2.0M	Assumption

All three cases pay back within the first year, but for different reasons.

Hospitals gain from recovered revenue; the billing firm from staff time

Modeled first-year benefit and cost ($ millions); each panel has its own scale

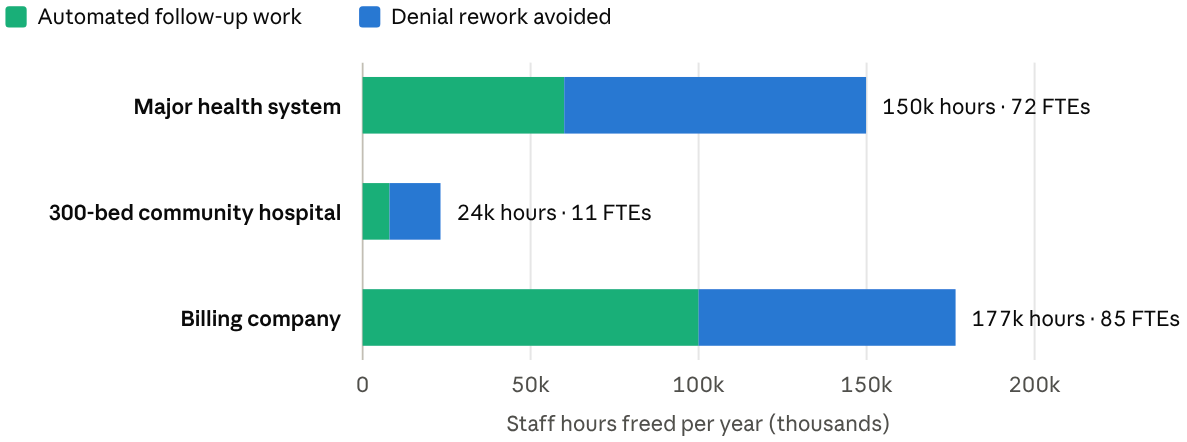


Illustrative model · 3 scenarios · inputs in the table above

Recovered revenue makes up 73–79% of each hospital's benefit, while staff time makes up 51% of the billing company's. Each panel has its own scale, so compare the shapes rather than bar lengths across panels.

AI frees the work of 11 to 85 full-time staff a year

Modeled staff hours freed per year, by source; 1 FTE = 2,080 hours



Illustrative model · 3 scenarios · rework hours derived from each scenario's rework cost and hourly rate

The hours come from two places: automated follow-up work and the rework on 60,000, 9,600 and 66,000 denials avoided a year. Rework hours assume each scenario's rework cost is staff time at its own hourly rate. Freed hours are capacity to redeploy to complex denials and audits, not automatic headcount cuts.

- **Major health system:** the largest prize in dollars, but it rests on a mature baseline. If AI cuts denials 10% rather than 15%, net benefit falls to about $14.0M. Integrating across many hospitals and EHR instances is the main cost risk.

- **300-bed community hospital:** a similar ~4x return on a far smaller base. The upfront spend and a thin IT team are the barriers, so EHR-native or vendor-hosted tools fit best. If the average written-off claim is $400 rather than $800, net falls to about $1.4M.
- **Billing company:** the lowest ratio, because it keeps only its 6% fee on revenue it recovers for clients. Its gain is labor productivity, worth about 4 points of margin on an assumed $90M in fee revenue ($1.5B collected). If only half the hours are freed, net falls to about $2.4M.

The billing company also faces two strategic risks the providers do not: clients will expect AI savings to be passed through as lower fees, and EHR-native AI such as Epic's Penny competes for the same work.

Governance essentials

- **Human accountability:** a named person signs off on every payer-facing output – codes above a risk threshold, prior-auth submissions and appeals.
- **Validation before go-live:** test each tool on a coded sample against current accuracy, then audit a fixed sample every month.
- **Vendor terms:** business associate agreement, limits on data use, audit rights, performance guarantees and a manual fallback plan.
- **Workforce plan:** retrain coders and billers as auditors, denial analysts and payer-policy specialists before cutting roles.
- **Payer intelligence:** use payers' new public prior-auth metrics to spot outlier denial patterns and bring them to contract talks.

KPIs to track

KPI	What it measures	Why it matters
Initial denial rate	Share of claims denied on first submission	The core outcome denial-prevention AI is bought to move
Clean claim rate	Share of claims accepted with no edits	Leading indicator of front-end data quality
Days in A/R	Average days from service to payment	Cash flow; captures coding and follow-up speed
DNFB days	Days discharged but not yet billed	Measures mid-cycle coding bottlenecks

KPI	What it measures	Why it matters
Touchless rate	Share of accounts handled with no human touch	Direct measure of automation depth
Coding accuracy	Audit agreement with certified coders	Guards against compliance risk
Prior-auth turnaround	Hours from order to decision	Patient access and scheduling
Appeal overturn rate	Share of appealed denials won	Shows whether AI-drafted appeals work
Cost to collect	Revenue-cycle cost per dollar collected	The bottom-line efficiency measure

Outlook to 2030 and conclusion

Through 2030, AI will likely become standard in hospital revenue cycles, but the biggest savings depend on shared payer–provider infrastructure rather than smarter algorithms on each side. Spending intent is strong: 70–90% of decision-makers expect to raise AI revenue-cycle budgets over the next three years (Oliver Wyman).

What to watch

- **January 2027 payer APIs.** Once FHIR prior-auth APIs are live, AI can move from filling portals to real-time, system-to-system approvals.
- **WISeR results.** Evidence from the six-state Medicare model will shape whether AI-assisted prior auth spreads across traditional Medicare.
- **The coding-intensity fight.** Expect payer policies, audits and possibly regulation aimed at AI-scribe-driven coding.
- **Agents inside the EHR.** EHR vendors are building AI agents directly into billing workflows, which should narrow the gap between large and small hospitals.
- **Market growth.** One estimate puts the U.S. revenue-cycle management market at $90.6 billion today, rising to $308 billion by 2030 (HFMA).

Conclusion

AI's impact on U.S. hospital revenue cycles in 2026 is best described as **targeted, measurable gains in specific workflows, not yet a change in the cost structure of health care**. Hospitals that start with rote tasks, measure against a baseline and keep humans accountable are seeing results. The industry's larger prize, much of the ~$18 billion a year spent fighting claims that end up paid, will come only when payers and providers use AI to prevent disputes rather than to win them.

Method and sources

This report synthesizes industry surveys, government sources, trade press and published case studies from 2021 to September 2026. Case-study results are as reported by the provider or vendor and have not been independently verified.

Industry data and surveys

1. Premier – Claims adjudication costs providers $25.7 billion (Feb 2025)
2. Experian Health – State of Claims survey 2025 (Sep 2025)
3. CAQH Index 2025 (Feb 2026)
4. McKinsey – Administrative simplification (Oct 2021)
5. Oliver Wyman – AI impact on revenue cycle (Apr 2026)
6. Fierce Healthcare – HFMA–AKASA survey on AI adoption (Dec 2025)
7. Healthcare Finance News – HFMA survey of finance leaders (May 2026)
8. HFMA – The revenue cycle of the future (2026)
9. AMA – Prior authorization physician survey (May 2026)
10. KFF – Medicare Advantage prior authorization, 2024 (Jan 2026)

Case studies

11. AHA – 3 ways AI can improve revenue-cycle management (Jun 2024)
12. Epic – Real results from Epic AI (Mar 2026)
13. Becker's – Autonomous coding at Mercyhealth (Jun 2025)

Payers and the AI arms race

14. Healthcare Finance News – The AI arms race in the revenue cycle (May 2026)
15. Healthcare Brew – The AI arms race between insurers and providers (Oct 2025)
16. TechTarget – How AI scribes are shifting coding intensity (Mar 2026)

17. STAT – Insurers and providers agree AI scribes raise bills (Apr 2026)
18. AHA – Fact sheet: AI and coding intensity (Jul 2026)
19. Healthcare Finance News – UnitedHealth AI class action advances (Feb 2025)
20. Fierce Healthcare – Insurers cut 11% of prior authorizations (Apr 2026)

Regulation and risk

21. CMS – Interoperability and Prior Authorization Final Rule (CMS-0057-F)
22. CareEvolution – Understanding CMS-0057-F
23. KFF – Regulation of AI in prior authorization and claims review (May 2026)
24. KFF – Potential impact of Medicare's WISeR model
25. Becker's – The AI arms race in utilization management (Apr 2026)
26. HHS – Change Healthcare cybersecurity incident FAQ (updated Jul 2025)

Cost model inputs

27. Tebra – How much should medical billing services charge? (updated Nov 2025)
28. BLS – Medical records specialists, Occupational Outlook Handbook (May 2025 wage data)

PORTFOLIO CASE STUDY · HEALTHCARE ADMINISTRATION

AI in the U.S. hospital revenue cycle

How automation is changing claims, denials and prior authorization, and where hospitals should start

Neha Vittal

September 2026

THE BOTTOM LINE

AI is delivering targeted wins, not yet a new cost structure

80%

of health systems are exploring, piloting or implementing generative AI for the revenue cycle

27%

run AI at scale across several revenue-cycle functions

18–71%

gains on single workflows in published case studies

Recommendation: start with rote, measurable work, and keep a person accountable for every payer-facing decision.

Sources: HFMA–AKASA via Fierce Healthcare (2025); HFMA (2026); AHA, Epic, Becker's case studies

THE PROBLEM

Fighting denials cost providers $25.7 billion in 2023

~15%

of private-payer claims are denied on first submission

~70%

of those denials are eventually paid, after costly rework

$57.23

average provider cost to rework one denied claim

13 hrs

a week of physician and staff time spent on prior authorization

Most denials are reversed, so the cost is **rework**, the kind of waste automation targets.

Sources: Premier (Feb 2025), claims data for 2023; AMA physician survey (2026)

WHERE AI IS APPLIED

AI now touches every stage of the revenue cycle

Front end

- Eligibility and coverage discovery
- Prior authorization
- Patient cost estimates

Mid-cycle

- Ambient documentation
- Clinical documentation improvement
- Assisted and autonomous coding

Back end

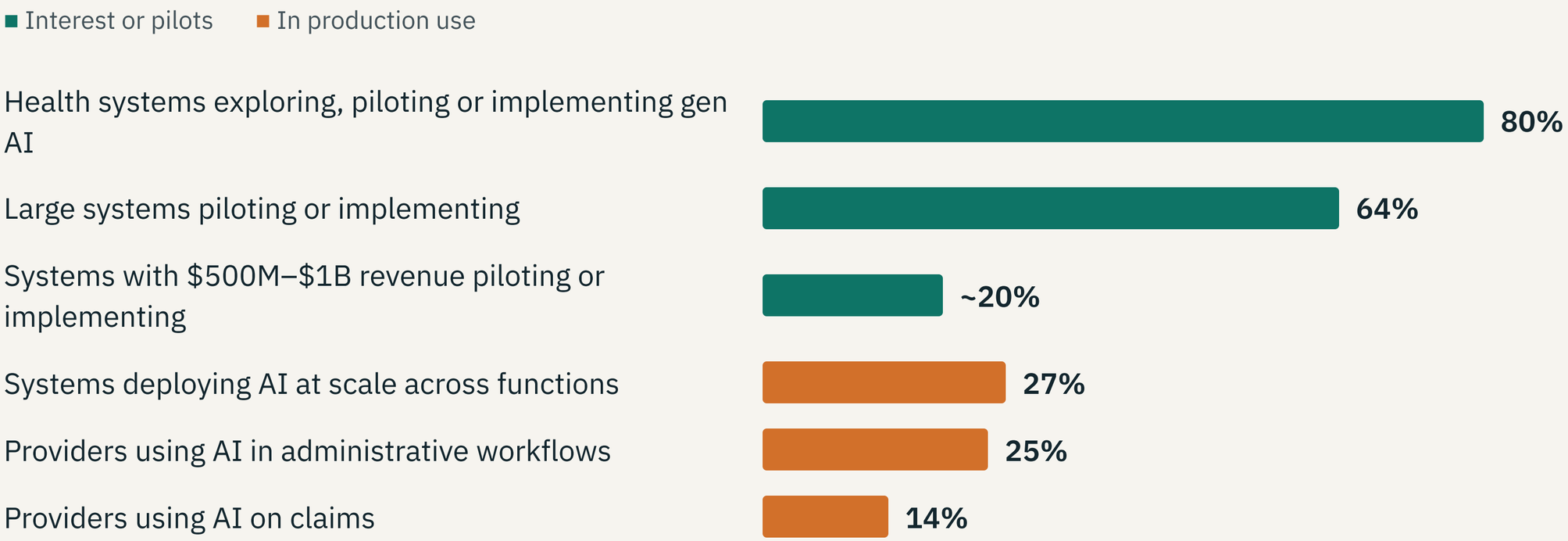
- Claim scrubbing and denial prediction
- Claim-status follow-up
- Appeal drafting
- Patient billing service

92% of leaders call some AI bets "no-regret", led by documentation, coding and e-prior auth.

Source: Oliver Wyman 2026 revenue-cycle survey

ADOPTION

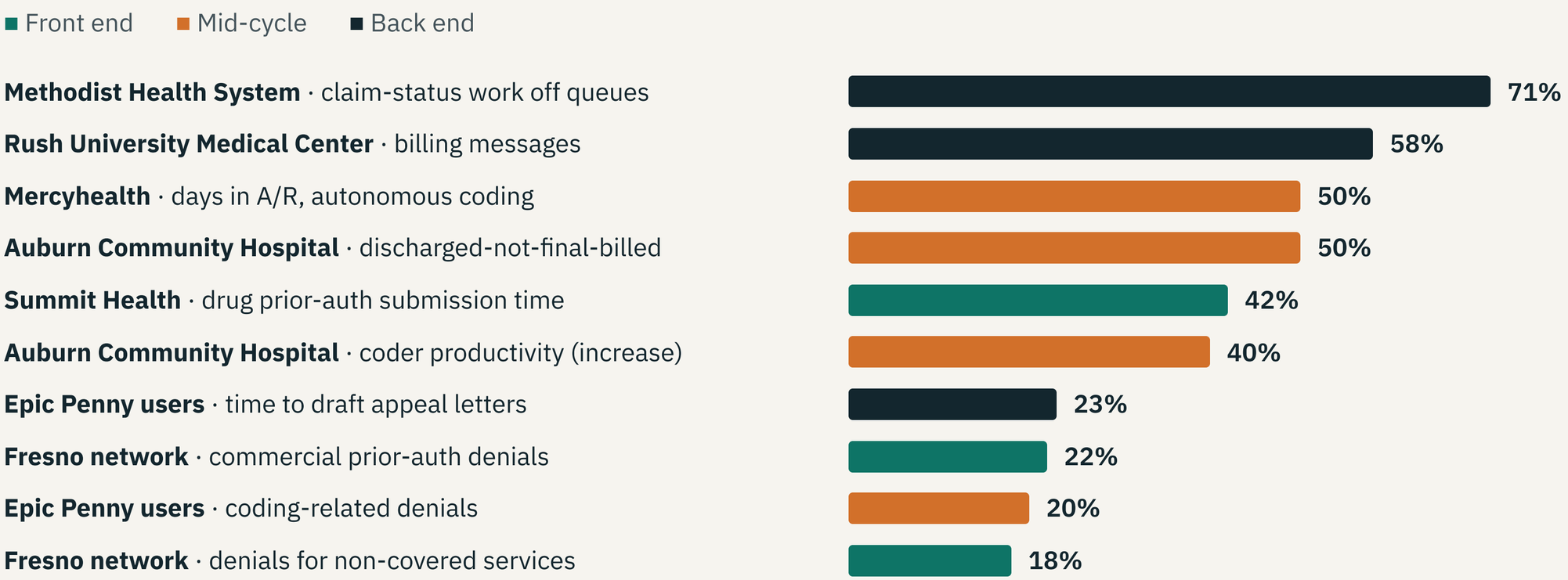
Interest is near-universal; production use is not



Sources: HFMA–AKASA via Fierce Healthcare (2025, n=519); HFMA (2026, n=95); CAQH Index (2026); Experian Health (2025). Surveys differ; compare direction, not decimals.

EVIDENCE OF IMPACT

Case studies report 18–71% gains on single workflows



Provider- or vendor-reported, not independently audited. Sources: AHA (2024), Epic (2026), Fierce Healthcare (2025), Becker's (2025)

THE PAYER SIDE

Insurers are automating faster than hospitals

50%+ vs 25%

of health plans vs providers use AI in administrative workflows

84%

of 93 insurers surveyed use AI or machine learning

$2.3B

added spending that one payer analysis attributes to AI scribes

An AI arms race on four fronts

Front	Payer AI	Provider response
Prior auth	Automated approvals; Aetna: 80% in real time	AI builds and tracks requests
Coding	Automated downcoding edits	Scribes, CDI, autonomous coding
Adjudication	Every claim screened before payment	Claim scrubbing, denial prediction
Appeals	Care-length algorithms (alleged in lawsuits)	Auto-drafted appeal letters

Sources: CAQH Index (2026); NAIC via KFF (2026); Blue Health Intelligence via TechTarget (2026); Healthcare Finance News (2026)

REGULATION

Rules now push toward faster, human-accountable decisions

2024

CMS: Medicare Advantage denials cannot rest on an algorithm alone

Jan 2026

CMS-0057-F: prior-auth decisions in 72 hours (urgent) or 7 days

Jan 2026

WISeR: AI-assisted prior auth reaches traditional Medicare in 6 states

Apr 2026

At least 8 states bar algorithm-only denials

Jan 2027

Payers must run FHIR prior-auth APIs, enabling real-time approvals

So what: payers now publish approval, denial and turnaround metrics (from March 2026). Hospitals can use them to spot outliers and bring them to contract talks.

Sources: CMS; CareEvolution; KFF state tracker (May 2026); KFF on WISeR

RISKS

Six risks every hospital must manage

Errors at scale

One wrong rule can hit thousands of claims; 25% of finance leaders fear exactly this.

Compliance exposure

AI-raised codes without documentation invite clawbacks and False Claims Act risk.

Automation bias

Reviewers anchor on AI output; no studies yet compare wrongful-denial rates.

Vendor and cyber risk

The Change Healthcare attack exposed ~192.7M people and froze claims nationwide.

Workforce

Only 7.4% of leaders call staff "very prepared"; aim to elevate, not eliminate.

Hospital equity

Large systems pilot AI about three times as often as smaller ones.

Sources: HFMA (2026); Stanford review in Health Affairs via Becker's (2026); HHS OCR (2025); Fierce Healthcare (2025)

RECOMMENDED ROADMAP

Start with rote work; earn the right to autonomy

Phase 1 · Months 0–6

Automate rote work

- Claim-status follow-up
- Eligibility checks
- Claim scrubbing

Goal: fund the next phase

Gate 1

Savings exceed cost

Phase 2 · Months 6–18

Assist skilled staff

- Coding assistance and CDI
- Prior-auth packets
- Appeal-letter drafting

Goal: raise output per FTE

Gate 2

Coding audit passes

Phase 3 · Months 18–36

Scale autonomy

- Autonomous coding
- API-based prior auth
- Patient billing assistant

Goal: lower cost to collect

Author's recommendation, based on tool maturity and the January 2027 payer API deadline

ILLUSTRATIVE BUSINESS CASES

All three pay back in year one, but for different reasons

Value mix: ■ Rework avoided ■ Recovered revenue ■ Staff time freed

Major health system

4M claims · 10% denials · AI cuts 15% · $5.0M cost

$22.3M

net benefit, **4.5x** its cost



79% from recovered revenue

300-bed community hospital

400K claims · 12% denials · AI cuts 20% · $0.6M cost

$2.5M

net benefit, **4.2x** its cost



73% from recovered revenue

Billing company

3M claims · 11% denials · keeps a 6% fee · $2.0M cost

$3.9M

net benefit, **2.0x** its cost



51% from staff time

Illustrative model; every input is an assumption to replace with local data. Inputs on slide 13.

TIME SAVINGS

AI frees the work of 11 to 85 full-time staff a year

Staff hours freed per year: ■ Automated follow-up work ■ Denial rework avoided



Elevate, not eliminate: freed hours are capacity to move to complex denials, audits and patient financial help, not automatic job cuts.

Illustrative model. Rework hours = denials avoided × rework cost ÷ hourly rate; 1 FTE = 2,080 hours. Inputs on slide 13.

MODEL INPUTS

The assumptions behind each case

Input	Major health system	300-bed hospital	Billing company
Claims per year	4,000,000	400,000	3,000,000
Initial denial rate	10%	12%	11%
Reduction in denials from AI	15%	20%	20%
Rework cost per denied claim	$57.23	$57.23	$35
Denials never recovered	30%	30%	30%
Average written-off claim	$1,200	$800	$500
Share of recovered revenue kept	100%	100%	6% fee
Staff hours freed per year	60,000	8,000	100,000
Loaded cost per staff hour	$38	$35	$30
Annual AI cost, incl. integration	$5.0M	$0.6M	$2.0M

Bases: Premier (2025) rework cost and recovery rate; Tebra (2025) billing fees of 4–10% of collections; BLS (May 2025) median wage of $24.59/hr. Others assumed.

CONCLUSION

The bigger prize is using AI to prevent payer–provider disputes, not to win them.

What to watch

January 2027 payer APIs make real-time prior auth possible

WISeR results decide whether AI prior auth spreads in Medicare

The AI-scribe coding-intensity fight shapes payer policy

Full report with sources: AI in U.S. Hospital Revenue Cycle Management (Neha Vittal, 2026)

SOURCES

Key sources

- Premier: Claims adjudication costs (Feb 2025)
- Experian Health: State of Claims (Sep 2025)
- CAQH Index 2025 (Feb 2026)
- Oliver Wyman: AI in revenue cycle (Apr 2026)
- Fierce Healthcare: HFMA–AKASA survey (Dec 2025)
- Healthcare Finance News: HFMA survey (May 2026)
- AMA: Prior authorization survey (May 2026)
- AHA: 3 ways AI can improve RCM (Jun 2024)
- Tebra: Medical billing service fees (Nov 2025)
- Epic: Real results from Epic AI (Mar 2026)
- Becker's: Mercyhealth autonomous coding (2025)
- Healthcare Finance News: AI arms race (May 2026)
- TechTarget: AI scribes and coding (Mar 2026)
- KFF: Regulating AI in prior auth (May 2026)
- KFF: Medicare's WISeR model
- CareEvolution: Understanding CMS-0057-F
- HHS: Change Healthcare incident FAQ
- BLS: Medical records specialists pay (May 2025)

Full list of 28 sources in the accompanying report

APPENDIX

Resume bullets

Figures are framed as modeled or projected, not as savings delivered.

Full version

AI in U.S. Hospital Revenue Cycle Management — Independent research project, Sep 2026

- Researched and wrote a report and a 15-slide executive deck on how AI is changing U.S. hospital revenue cycle management, drawing on 28 industry, government and trade sources (Premier, HFMA, CAQH, KFF, CMS).
- Sized the claim-denial problem: $25.7B a year in provider rework costs, with ~15% of claims denied at first and ~70% of those later paid. Mapped AI use cases across front-end, mid-cycle and back-end workflows.
- Benchmarked published results from 7 health systems and platforms, which showed 18–71% gains on single workflows. Also pointed out the limits of results reported by vendors and providers themselves.
- Built a three-scenario ROI model (major health system, 300-bed community hospital, billing company) with sensitivity analysis. It projected $2.5M–$22.3M in first-year net benefit and 24k–177k staff hours freed a year (11–85 full-time staff).
- Analyzed insurers' use of AI and new regulation (CMS-0057-F, Medicare's WISeR model, state laws on AI-driven denials) to assess compliance, automation-bias and vendor-concentration risks.
- Designed a phased rollout plan with go/no-go checkpoints, governance controls and 9 KPIs, including denial rate, days in A/R and cost to collect.

Short version

- Wrote a report and executive deck on AI in U.S. hospital revenue cycles, drawing on 28 sources, including a review of 7 organizations' published results showing 18–71% workflow gains.
- Built a three-scenario ROI model projecting $2.5M–$22.3M in first-year net benefit and 11–85 full-time staff's worth of hours freed, with sensitivity analysis.
- Assessed insurer AI, CMS and state regulation, and operating risks, and recommended a phased rollout plan with 9 KPIs.